DBMS Tables Normalization Lab

Question 1

Problem Statement:

Normalize the table up to 3NF.

Dataset:

OrderID	CustomerName	Product	Price
1	Anuj	Laptop	50000
1	Anuj	Mouse	500
2	Amit	Laptop	50000
2	Amit	Keyboard	1500
3	Ravi	Mouse	500
4	Sita	Laptop	50000
5	Mohan	Keyboard	1500
6	Riya	Mouse	500
7	Karan	Laptop	50000
8	Neha	Keyboard	1500

Question 2

Problem Statement:

Identify dependencies and normalize to 3NF.

Dataset:

EmpID	EmpName	DeptID	DeptName
101	Anuj	D1	HR
102	Amit	D2	IT
103	Ravi	D1	HR
104	Sita	D3	Finance
105	Mohan	D2	IT
106	Riya	D1	HR
107	Karan	D3	Finance
108	Neha	D2	IT
109	Pooja	D1	HR
110	Rahul	D3	Finance

Question 3

Problem Statement:

Remove transitive dependency and normalize.

Dataset:

StudentID	Subject	Teacher	TeacherPhone
1	Math	Ravi	9999
1	Science	Ravi	9999
2	English	Sita	8888
3	Math	Ravi	9999
4	Science	Ravi	9999
5	English	Sita	8888
6	Math	Ravi	9999
7	Science	Ravi	9999
8	English	Sita	8888
9	Math	Ravi	9999

Question 4

Problem Statement:

Normalize to 3NF and identify all dependencies.

Dataset:

BookID	Author	Publisher	PublisherAddress
B1	A1	P1	Delhi
B2	A2	P2	Mumbai
B3	A1	P1	Delhi
B4	A3	P3	Pune
B5	A2	P2	Mumbai
B6	A1	P1	Delhi
B7	A3	P3	Pune
B8	A2	P2	Mumbai
B9	A1	P1	Delhi
B10	A3	P3	Pune

Question 5

Problem Statement:

Find candidate key and normalize to 3NF.

Dataset:

OrderID	ProductID	ProductName	CustomerID	CustomerName
1	P1	Laptop	C1	Anuj
1	P2	Mouse	C1	Anuj
2	P1	Laptop	C2	Amit
2	P3	Keyboard	C2	Amit
3	P2	Mouse	C3	Ravi
4	P1	Laptop	C4	Sita
5	P3	Keyboard	C5	Mohan
6	P2	Mouse	C6	Riya
7	P1	Laptop	C7	Karan
8	P3	Keyboard	C8	Neha

Question 6

Problem Statement:

Normalize the student-course table up to 3NF.

Dataset:

StudentID	StudentName	Course	Instructor
1	Anuj	Math	Ravi
1	Anuj	Science	Ravi
2	Amit	English	Sita
3	Ravi	Math	Ravi
4	Sita	Science	Ravi
5	Mohan	English	Sita
6	Riya	Math	Ravi
7	Karan	Science	Ravi
8	Neha	English	Sita
9	Pooja	Math	Ravi

Question 7

Problem Statement:

Normalize employee-project data to 3NF.

Dataset:

EmpID	ProjectID	ProjectName	Manager
101	P1	Alpha	Raj
101	P2	Beta	Raj
102	P1	Alpha	Raj
103	P3	Gamma	Simran
104	P2	Beta	Raj
105	P3	Gamma	Simran
106	P1	Alpha	Raj
107	P2	Beta	Raj
108	P3	Gamma	Simran
109	P1	Alpha	Raj

Question 8

Problem Statement:

Normalize sales data up to 3NF.

Dataset:

SaleID	Product	Category	Price
1	Laptop	Electronics	50000
2	Mouse	Electronics	500
3	Keyboard	Electronics	1500
4	Shirt	Clothing	1000
5	Jeans	Clothing	2000
6	Laptop	Electronics	50000
7	Mouse	Electronics	500
8	Shirt	Clothing	1000
9	Jeans	Clothing	2000
10	Keyboard	Electronics	1500

Question 9

Problem Statement:

Normalize hospital patient records to 3NF.

Dataset:

PatientID	PatientName	Doctor	DoctorSpecialization
1	Anuj	Dr. Ravi	Cardiology
2	Amit	Dr. Sita	Neurology
3	Ravi	Dr. Ravi	Cardiology
4	Sita	Dr. Sita	Neurology
5	Mohan	Dr. Ravi	Cardiology
6	Riya	Dr. Sita	Neurology
7	Karan	Dr. Ravi	Cardiology
8	Neha	Dr. Sita	Neurology
9	Pooja	Dr. Ravi	Cardiology
10	Rahul	Dr. Sita	Neurology

Question 10

Problem Statement:

Normalize university data to 3NF.

Dataset:

StudentID	Dept	HOD
1	CSE	Dr. A
2	IT	Dr. B
3	CSE	Dr. A
4	ECE	Dr. C
5	IT	Dr. B
6	CSE	Dr. A
7	ECE	Dr. C
8	IT	Dr. B
9	CSE	Dr. A
10	ECE	Dr. C